Dhruva Chayapathy

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Education

Georgia Institute of Technology Atlanta, GA
Bachelors in Computer Engineering *Expected May 2028*
GPA: 3.8

Threads: Distributed Systems & Software Design, Systems and Architecture
Awards: HackGT 2nd Overall (MLH Hackathon), MIT iQuHack 2nd in QuEra Track, Faculty Honors
Publications: First Author 2x NeurIPS Workshops, Coauthor 1x ICLR Workshop, First Author 1x Springer AI, Coauthor 1x Journal of Fusion Energy

Experience

Software Development Engineering Intern | *Java, TypeScript, AWS* May 2026 – August 2026
Amazon Web Services – AWS Config/Amazon Dedicated Cloud

- Built a platform to migrate, deploy, and manage **300,000+** production AWS Config alarms/monitors.
- Developed an ETL pipeline converting raw, messy legacy alarm data into simplified meta-alarm configs.
- Engineered team-specific **no-op** code-gen and CDK pipelines deploying their regional CloudWatch alarms.
- Cut deployments from **13-hour** legacy cron jobs to **under 10 minutes** for **10+** AWS Config teams.

Research Assistant | *Python, PyTorch, JAX, Dask* August 2024 – December 2025
MIT Plasma Science and Fusion Center

- Performed data analysis/exploration and PCA on **300M** rows of tokamak disruption data using **Dask**.
- Trained PyTorch deep learning models for time-series disruption prediction, achieving **0.881 ROC-AUC**.
- Conducted research on generative models for augmentation of disruption datasets, enlarging by over **3x**.
- Showed inclusion of synthetic data resulted in **18.1%** improvement in F2 for low-data regimes.

Research Assistant | *Python, PyTorch, PyTorch Geometric, HuggingFace* August 2024 – December 2025
Georgia Tech Fung Group

- Optimized MatDeepLearn GNN pipelines, cutting **memory usage 50%** with negligible performance loss.
- Increased predictive performance of GNNs by **11.5%** through feature engineering of crystal graph embeddings.
- Worked on LLM4StructGen, benchmarking 2 LLMs, 4 text representations, and 2 preprocessing configurations.

Projects

dInference: Verifiable LLM Inference on Solana | *Python, Rust, PyTorch, Anchor, Solana, Walrus*

- Designed a permissionless Solana protocol for verifiable LLM inference on open-source models.
- Built a PyTorch hidden-state fingerprint to verify a response came from the claimed model, with just **256 bytes**.
- Detected model substitution at **100% TPR / 0% FPR** over **1,500 samples**, incl. same-family downsizing.
- Cut verification to one forward pass, **∼18x** faster than re-execution, with **1.3%** provider overhead.

Muse Spark GPU Kernel Optimization | *Python, C++, CUDA, Triton, PyTorch*

- Optimized projection kernels for Muse Spark, an **816B** MoE, to **1.35x** SGLang's production kernels on H200.
- Implemented a lossless **12-bit** compression and CUDA kernel for skinny matrices, **1.5–1.8x** over cuBLAS kernels.
- Wrote fused FP32-exact routing kernels, **2–4x** faster than SGLang, tuned per shape via Bayesian search.

UtterDB: Distributed Key-Value Database | *Golang, Terraform, gRPC, Locust*

- Built a distributed key-value store in Go achieving **100,000+ req/s** at **1ms end-to-end p50 latency**.
- Implemented a **Terraform**-provisioned, multi-node cluster to facilitate horizontal scaling of data and routing.
- Engineered a control plane with heartbeating, replica failovers, and **zero-downtime** rebalancing.
- Stress-tested using Locust, with over **150 million requests**, and **0 read/write failures.**

Technical Skills

Languages: Python, Golang, Java, C++, TypeScript, Rust, SQL
Systems & Infrastructure: AWS (EC2, S3, Lambda, CDK, and more), Docker, Linux
Databases & Data: PostgreSQL, Redis, Dask, Pandas, NumPy
Machine Learning: PyTorch, CUDA, Triton, JAX, HuggingFace, TensorFlow
Web & Tools: React, Next.js, Django, Flask, FastAPI, Hono, Git, Vercel